



LOCAL AGENTIC AI

Manual Startup, Configuration & Troubleshooting Guide

100% LOCAL INFERENCE

React Frontend

http://localhost:3000 ⇌ WebSocket /
HTTP ⇌

FastAPI Backend

http://localhost:8000 ⇌
LangChain ⇌

Existing Agent

agent.py +
tools.py

⇌ Local
API ⇌

Ollama Daemon

qwen3:4b (Port
11434)

⚡ 1. Manual Startup Commands (Run in 3 Terminals)

Terminal 1 – Local Model Daemon (Ollama)

Port 11434

```
# Starts the Ollama server hosting qwen3:4b locally
ollama serve
```

Terminal 2 – FastAPI Backend Server

Port 8000

```
# Navigate to project (note the quote for space in folder name)
cd "/Users/maxe/Desktop/Mini_project /AI_agent/agent-ai"

# Activate virtual environment
source .venv/bin/activate

# Start the FastAPI server with live reload
uvicorn backend.server:app --reload --port 8000
```

Terminal 3 – React Dashboard (Vite)

Port 3000

```
# Navigate to frontend
cd "/Users/maxe/Desktop/Mini_project /AI_agent/agent-ai/frontend"

# Start development server
npm run dev
```

Optional – Original Terminal CLI Mode (No Browser Needed)

CLI

```
cd "/Users/maxe/Desktop/Mini_project /AI_agent/agent-ai"
source .venv/bin/activate
python main.py
```

🔧 2. Expected Problems & How to Overcome Them

Problem / Error	Root Cause	How to Overcome (Solution)
Port 8000 or 3000 in Use Errno 48: Address already in use	A previous background instance of Uvicorn or Vite is still running and occupying the port.	Kill the process holding the port: <code>lsof -ti:8000 xargs kill -9</code> <code>lsof -ti:3000 xargs kill -9</code> Then re-run the startup command.
🔴 Ollama Offline Connection refused (port 11434)	Ollama service is stopped, macOS restarted, or the daemon crashed.	1. Verify status: <code>curl http://localhost:11434/api/tags</code> 2. If down, restart it: <code>pkill ollama && ollama serve</code> 3. Reload the browser page.
Model Not Found model 'qwen3:4b' not found	The Qwen3 4B GGUF weights are not installed in your local Ollama store.	Pull the model locally: <code>ollama pull qwen3:4b</code> Verify with: <code>ollama list</code>
ModuleNotFoundError No module named 'fastapi'	The command was executed using system Python instead of the virtual environment.	Activate the virtual environment first: <code>source .venv/bin/activate</code> Or run directly using: <code>.venv/bin/uvicorn backend.server:app --port 8000</code>
Vite Command Not Found sh: vite: command not found	Node dependencies were not installed in the <code>frontend/</code> folder.	Install frontend dependencies: <code>cd frontend && npm install</code>
Tool Timeout (Weather / Search) urlopen error / search failed	<code>get_weather</code> and <code>web_search</code> require internet access.	The calculator and file tools are 100% offline. Ensure your Mac has an internet connection if testing weather or DuckDuckGo searches.
Path Space Syntax Error cd: no such file or directory	The parent folder contains a trailing space: <code>Mini_project /</code> .	Always wrap folder paths in quotes: <code>cd "/Users/maxe/Desktop/Mini_project /AI_agent/agentica-ai"</code>

🔍 3. Quick System Health Checks

Quick Verification Commands

Bash / Zsh

```
# 1. Test Ollama model tags
curl -s http://localhost:11434/api/tags
```

```
# 2. Test FastAPI health endpoint
curl -s http://localhost:8000/api/health

# 3. Test Tools registry endpoint
curl -s http://localhost:8000/api/tools

# 4. Open Dashboard in Default Browser
open http://localhost:3000
```